

Unitree Week 22

Presentation by Mario Ortiz

This Week

- More work on Unitree R1 Reinforcement Learning code
- More work on Inverse Kinematics Code

First RL Attempt with R1:

- After implementing entire G1's `rough_env_cfg.py` onto an R1 integration, I attempted to train the R1 to walk on flat terrains.
 - The G1 code utilizes “torso_link” to cause termination penalties to punish the robot for falling onto the ground.
 - The R1 doesn't have this link, and instead has a “pelvis_link”, which causes an R1 “episode” to not terminate when it falls onto the ground.
 - Solution: Ensure that any R1 upper body part contacting the ground gets penalized (maybe ensure that anything other than the feet/lower legs touching the ground causes episode termination).

First RL Attempt with R1:

- This can be seen in the reward section, where the robot achieves the maximum possible mean episode length of 1000, as well as achieve a mean reward of 17.03. This is achieved regardless of the fact that the robot instantly falls onto the ground.
- This is due to the robot causing the R1 to learn to fall onto the ground without allowing this “pelvis_link” to contact the ground.

```
#####
Learning iteration 2999/3000

      Total steps: 73728000
      Steps per second: 7521
      Collection time: 3.009s
      Learning time: 0.259s
      Mean value loss: 0.0005
      Mean surrogate loss: -0.0072
      Mean entropy loss: 7.8399
      Mean reward: 17.03
      Mean episode length: 1000.00
      Mean action std: 0.43
Episode_Reward/track_lin_vel_xy_exp: 1.7870
Episode_Reward/track_ang_vel_z_exp: 0.3873
Episode_Reward/lin_vel_z_l2: 0.0000
Episode_Reward/ang_vel_xy_l2: -0.0323
Episode_Reward/dof_torques_l2: -0.0012
Episode_Reward/dof_acc_l2: -0.0098
Episode_Reward/action_rate_l2: -0.0988
Episode_Reward/feet_air_time: 0.0002
Episode_Reward/flat_orientation_l2: -0.9544
Episode_Reward/dof_pos_limits: -0.0057
Episode_Reward/termination_penalty: 0.0000
Episode_Reward/feet_slide: -0.0034
Episode_Reward/joint_deviation_hip: -0.0362
Episode_Reward/joint_deviation_waist: -0.0348
Episode_Reward/joint_deviation_arms: -0.1244
Episode_Reward/joint_deviation_head: -0.0120
Metrics/base_velocity/error_vel_xy: 1.7525
Metrics/base_velocity/error_vel_yaw: 6.0268
Episode_Termination/time_out: 0.9980
Episode_Termination/base_contact: 0.0020
-----
Iteration time: 3.27s
Time elapsed: 02:10:59
ETA: 00:00:00
```

Second RL Attempt with R1:

- This attempt had initially appeared to solve the termination penalty issue, but we saw that in this attempt the R1 learned to maneuver without walking, and instead learned to hop.
- This was due to parameters such as:
 - vertical velocity penalty set to 0
 - `lin_vel_z_l2` weight = 0.0
 - Forward motion weights were high
 - `track_lin_vel_xy_exp` weight = 2.0

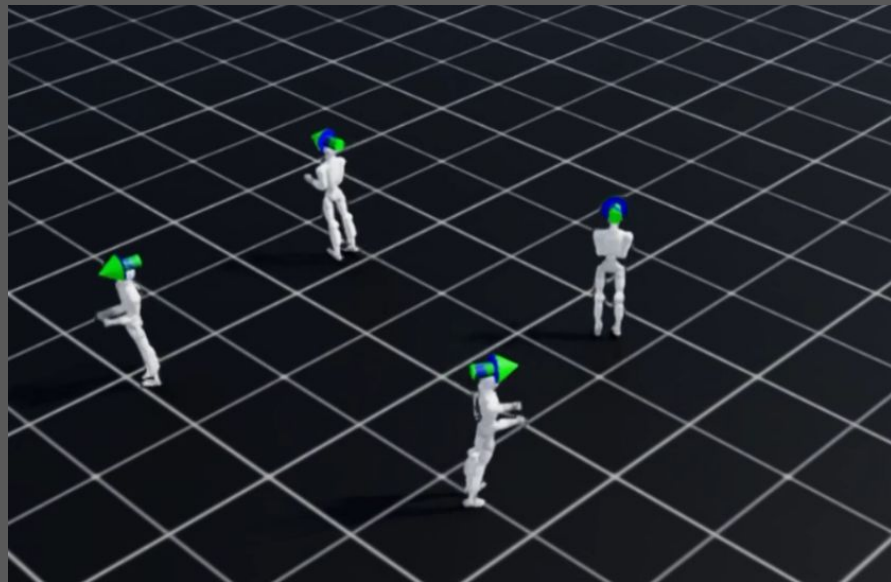
Third RL Attempt with R1:

- The next step now is to fix the reward/penalty parameters to ensure the robot is penalized for not using lower body parameters.
- This training here shows majority of the robots obtain a high reward without terminating (after implementing strict termination penalties).

```
#####  
Learning iteration 2999/3000  
  
Total steps: 73728000  
Steps per second: 10044  
Collection time: 2.210s  
Learning time: 0.236s  
Mean value loss: 0.0013  
Mean surrogate loss: -0.0073  
Mean entropy loss: 11.3532  
Mean reward: 21.34  
Mean episode length: 964.79  
Mean action std: 0.57  
Episode_Reward/track_lin_vel_xy_exp: 0.9088  
Episode_Reward/track_ang_vel_z_exp: 0.5620  
Episode_Reward/lin_vel_z_l2: -0.0060  
Episode_Reward/ang_vel_xy_l2: -0.0314  
Episode_Reward/dof_torques_l2: -0.0013  
Episode_Reward/dof_acc_l2: -0.0046  
Episode_Reward/action_rate_l2: -0.1166  
Episode_Reward/feet_air_time: 0.0049  
Episode_Reward/flat_orientation_l2: -0.0068  
Episode_Reward/dof_pos_limits: -0.0166  
Episode_Reward/termination_penalty: -0.0104  
Episode_Reward/feet_slide: -0.0168  
Episode_Reward/joint_deviation_hip: -0.0171  
Episode_Reward/joint_deviation_waist: -0.0097  
Episode_Reward/joint_deviation_arms: -0.1257  
Episode_Reward/joint_deviation_head: -0.0225  
Metrics/base_velocity/error_vel_xy: 0.2217  
Metrics/base_velocity/error_vel_yaw: 0.7932  
Episode_Termination/time_out: 0.9832  
Episode_Termination/base_contact: 0.0168  
  
-----  
Iteration time: 2.45s  
Time elapsed: 02:08:41  
ETA: 00:00:00
```

Third RL Attempt with R1:

- The robots learn to properly walk in the flat environment (similarly to the G1's `flat_env_cfg.py`) after the 3000 iterations.
- They utilize small leg motion to achieve walking.



Commands:

Flat env cfg.py

Train:

```
.\isaacclab.bat -p .\scripts\reinforcement_learning\rsl_rl\train.py --task Isaac-Velocity-Flat-R1-v0  
--num_envs 1024 --headless
```

Play:

```
.\isaacclab.bat -p .\scripts\reinforcement_learning\rsl_rl\play.py --task  
Isaac-Velocity-Flat-R1-Play-v0 --num_envs 4
```

Rough env cfg.py

Train:

```
.\isaacclab.bat -p .\scripts\reinforcement_learning\rsl_rl\train.py --task Isaac-Velocity-Rough-R1-v0  
--num_envs 1024 --headless
```

Play:

```
.\isaacclab.bat -p .\scripts\reinforcement_learning\rsl_rl\play.py --task  
Isaac-Velocity-Rough-R1-Play-v0 --num_envs 4
```

Rough_env_cfg.py Test:

Rough_env_cfg.py is altered to work similar to the G1's version. The main changes were:

- The G1's rough_env_cfg.py utilizes randomized terrains; this was changed in the R1 version so there is only one terrain (rough terrain).
 - code
.\IsaacLab\source\isaacclab\isaacclab\terrains\config\rough.py
- Added penalties to punish R1 for wider stances

```
#####
Learning iteration 6999/7000

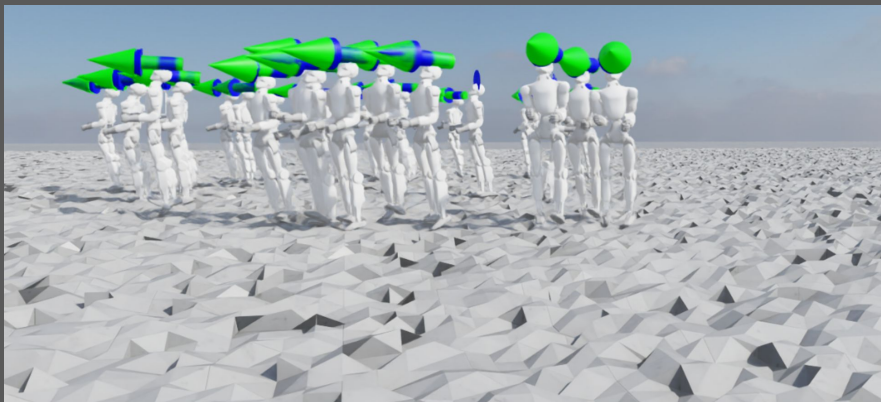
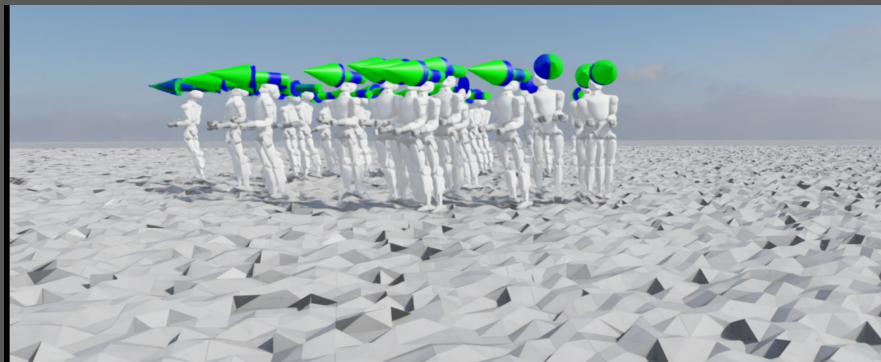
Total steps: 172032000
Steps per second: 8400
Collection time: 2.680s
Learning time: 0.246s
Mean value loss: 0.0225
Mean surrogate loss: -0.0067
Mean entropy loss: 35.0673
Mean reward: -0.41
Mean episode length: 851.23
Mean action std: 1.25
Episode_Reward/track_lin_vel_xy_exp: 0.6603
Episode_Reward/track_ang_vel_z_exp: 0.3696
Episode_Reward/lin_vel_z_l2: 0.0000
Episode_Reward/ang_vel_xy_l2: -0.1129
Episode_Reward/dof_torques_l2: -0.0018
Episode_Reward/dof_acc_l2: -0.0113
Episode_Reward/action_rate_l2: -0.3898
Episode_Reward/feet_air_time: 0.0042
Episode_Reward/flat_orientation_l2: -0.0079
Episode_Reward/dof_pos_limits: -0.0491
Episode_Reward/termination_penalty: -0.0500
Episode_Reward/feet_slide: -0.0237
Episode_Reward/joint_deviation_hip_roll: -0.0765
Episode_Reward/joint_deviation_hip_yaw: -0.0352
Episode_Reward/joint_deviation_waist: -0.0157
Episode_Reward/joint_deviation_arms: -0.1946
Episode_Reward/joint_deviation_head: -0.0414
Curriculum/terrain_levels: 5.4845
Metrics/base_velocity/error_vel_xy: 0.4236
Metrics/base_velocity/error_vel_yaw: 2.2281
Episode_Termination/time_out: 0.7537
Episode_Termination/base_contact: 0.2463

-----
Iteration time: 2.93s
Time elapsed: 05:46:55
ETA: 00:00:00
```

Rough_env_cfg Attempt with R1:

The R1's are now able to walk on rougher terrains.

- Notes:
 - Since the R1 has head assets (unlike the G1), the head of the R1 is placed improperly. The head appears to be looking upwards.



Pyramid_env_cfg & Stair_env_cfg

- For the G1, I created RL algorithms to train the robots to walk on pyramid terrains (pyramid_env_cfg.py) and a set of stairs that increased in height as we go up stairs (stair_env_cfg.py)
- Next steps can be allocating testing these programs with the R1.

Imitation Learning

Researching Imitation Learning Pipelines

- Human2Humanoid: Github repository that discusses an AMASS retargeting pipeline for Teleoperation to Humanoid robot simulation.
- The steps they discuss for retargeting is:
 1. Write the robot's forward kinematics
 2. Fit the SMPL body shape to match the humanoid kinematics
 3. Retarget AMASS using corresponding keypoints between fitted SMPL and robot.

Researching Imitation Learning Pipelines

- IN the AMASS retargeting that I have worked on, I had taken AMASS keypoints and immediately attempted to retarget them to the Unitree G1.
- There needs to be some sort of “robot-specific SMPL shape-fitting step” which ensures retargeting is possible.

SMPL Shape-Fitting

This is where we create a robot-specific human skeleton.

- Instead of retargeting from a human-shaped skeleton, we must retarget from a R1-shaped skeleton.
- The fitted SMPL body will have approximate R1 joint parameters (ex. hip length), simplifying the retargeting phase

USD File Analysis

- The R1 has 40 links
- The R1 has 26 controllable joints: q trajectory will be in the shape (T, 26)
- R1 is much smaller than AMASS Human Joints
- R1 forward direction: +x; R1 left direction: +y; R1 up direction: +z
 - root_pos trajectory: (T, 3)
 - root_yaw: (T, 4)
- T = number of frames
-

extract_r1_skeleton.py

- Inputs the R1's USD file and outputs the skeleton file of the robot
 - .\r1\outputs\r1_skeleton_neutral.npz
 - .\r1\outputs\r1_skeleton_body_map.json
- Outputs Keypoint Positions of the R1

```
=====
Extracting R1 neutral skeleton
=====
pelvis      -> pelvis_link
torso       -> waist_yaw_link
left_hip    -> left_hip_pitch_link
right_hip   -> right_hip_pitch_link
left_knee   -> left_knee_link
right_knee  -> right_knee_link
left_foot   -> left_ankle_roll_link
right_foot  -> right_ankle_roll_link
left_shoulder -> left_shoulder_pitch_link
right_shoulder -> right_shoulder_pitch_link
left_elbow  -> left_elbow_link
right_elbow -> right_elbow_link
left_hand   -> left_wrist_roll_link
right_hand  -> right_wrist_roll_link
head        -> head_yaw_link

Saved skeleton NPZ:
C:\Users\Mario\IsaacLab\scripts\r1\outputs\r1_skeleton_neutral.npz

Saved body map JSON:
C:\Users\Mario\IsaacLab\scripts\r1\outputs\r1_skeleton_body_map.json

Keypoint positions:
pelvis      : [+0.0000, +0.0000, +0.7918]
torso       : [+0.0325, +0.0000, +0.8408]
left_hip    : [+0.0325, +0.0705, +0.7016]
right_hip   : [+0.0325, -0.0705, +0.7016]
left_knee   : [+0.0147, +0.1069, +0.4134]
right_knee  : [+0.0147, -0.1069, +0.4134]
left_foot   : [-0.0021, +0.0863, +0.1043]
right_foot  : [-0.0021, -0.0863, +0.1043]
left_shoulder : [+0.0325, +0.0857, +1.0383]
right_shoulder : [+0.0325, -0.0857, +1.0383]
left_elbow  : [+0.0487, +0.1686, +0.8345]
right_elbow : [+0.0487, -0.1686, +0.8345]
left_hand   : [+0.1609, +0.1386, +0.8228]
right_hand  : [+0.1609, -0.1386, +0.8228]
head        : [+0.0275, +0.0000, +1.2123]
```

prepare_amass_targets.py

- Loads the saved R1 skeleton file from “extract_r1_skeleton.py”
- Loads “SMPLH_MALE.pk1” body model through the installed smplx package
- Fits SMPLH body shape/scale to R1 proportions by optimizing **betas** and the global **scale**
- Applies the AMASS motion to the fitted R1-sized SMPLH body.
- Converts SMPLH coordinates into Isaac-style coordinates.
- Extracts target keypoints over time (root position, root yaw, ...)
- Creates local frame targets relative to root yaw
- Creates vector targets (ex. Arm direction, foot-forward direction)
- Estimates foot contact flags
- Saves target files to an .npz file for retargeting

prepare_amass_targets.py

```
loading R1 neutral skeleton:
C:\Users\Mario\IsaacLab\scripts\r1\outputs\r1_skeleton_neutral.npz

R1 target lengths:
hip_width           : 0.14093 m
shoulder_width      : 0.17138 m
left_upper_leg      : 0.29099 m
right_upper_leg     : 0.29099 m
left_lower_leg      : 0.31031 m
right_lower_leg     : 0.31031 m
left_upper_arm      : 0.22065 m
right_upper_arm     : 0.22065 m
left_forearm        : 0.11672 m
right_forearm       : 0.11672 m
pelvis_to_head      : 0.42140 m
pelvis_to_shoulder_mid : 0.24862 m

Loading SMPLH male model:
C:\Users\Mario\Documents\Homework\Internship Stuff\ViSunInternship\smplx\smplx\smplh\SMPLH_MALE.pkl

Imported real smplx package from:
C:\Users\Mario\IsaacLab\isaac_sim\kit\python\Lib\site-packages\smplx\_init_.py
WARNING: You are using a SMPL+H model, with only 10 shape coefficients.

-----
Fitting SMPLH male proportions to R1
-----
iter 0000 | loss=1.797895 | shape_loss=1.797895 | scale=0.6500 | beta_norm=0.0949
iter 0100 | loss=1.125506 | shape_loss=1.119544 | scale=0.6134 | beta_norm=7.7777
iter 0200 | loss=0.868365 | shape_loss=0.852631 | scale=0.6193 | beta_norm=12.5867
iter 0300 | loss=0.697959 | shape_loss=0.670243 | scale=0.6553 | beta_norm=16.6857
iter 0400 | loss=0.589534 | shape_loss=0.549782 | scale=0.7035 | beta_norm=19.9648
iter 0500 | loss=0.526992 | shape_loss=0.477552 | scale=0.7412 | beta_norm=22.2524
iter 0600 | loss=0.485914 | shape_loss=0.428455 | scale=0.7618 | beta_norm=23.9844
iter 0700 | loss=0.454660 | shape_loss=0.389080 | scale=0.7755 | beta_norm=25.6221
iter 0800 | loss=0.430277 | shape_loss=0.355759 | scale=0.7891 | beta_norm=27.3115
iter 0900 | loss=0.411520 | shape_loss=0.327318 | scale=0.8036 | beta_norm=29.0305
iter 1000 | loss=0.397446 | shape_loss=0.303176 | scale=0.8184 | beta_norm=30.7153
iter 1100 | loss=0.387214 | shape_loss=0.282933 | scale=0.8326 | beta_norm=32.3026
iter 1199 | loss=0.380098 | shape_loss=0.266384 | scale=0.8455 | beta_norm=33.7293

Final fitted lengths:
hip_width           : target=0.14093 | fit=0.13412 | err=-0.00681 | -4.83%
shoulder_width      : target=0.17138 | fit=0.17568 | err=+0.00430 | +2.51%
left_upper_leg      : target=0.29099 | fit=0.29849 | err=+0.00749 | +2.58%
right_upper_leg     : target=0.29099 | fit=0.29837 | err=+0.00737 | +2.53%
left_lower_leg      : target=0.31031 | fit=0.30252 | err=-0.00779 | -2.51%
right_lower_leg     : target=0.31031 | fit=0.30063 | err=-0.00968 | -3.12%
left_upper_arm      : target=0.22065 | fit=0.16668 | err=-0.05397 | -24.46%
right_upper_arm     : target=0.22065 | fit=0.16628 | err=-0.05436 | -24.64%
left_forearm        : target=0.11672 | fit=0.13525 | err=+0.01854 | +15.88%
right_forearm       : target=0.11672 | fit=0.16765 | err=+0.05093 | +43.64%
pelvis_to_head      : target=0.42140 | fit=0.39798 | err=-0.02342 | -5.56%
pelvis_to_shoulder_mid : target=0.24862 | fit=0.26772 | err=+0.01909 | +7.68%

Saved SMPLH shape fit:
C:\Users\Mario\IsaacLab\scripts\r1\outputs\r1_smplh_shape_fit.npz
```

retarget_amass_r1.py

- Loads the prepared amass target files (r1_amass_targets.npz)
- Loads the R1 USD as an “IsaacLab Articulation”
- Loads the R1 body map + joint map
- Solves inverse kinematics of amass target files frame by frame.
 - Optimizes R1’s root position and joint angles
 - Keeps root yaw from the prepared AMASS target file
 - Tracks target keypoints (feet, knees, hips, chest, head, arms)
 - Tracks vector directions (torso up, hip side, shoulder side, thigh/shin directions, foot-forward directions).
- Regularization is added to avoid improper joint poses.
- Temporal Smoothing is added to remove frame-by-frame jitter.
- Final R1 motion file is then saved to “r1_retargeted_motion.npz”

retarget_amass_r1.py

```
=====
Starting R1 IK retargeting
=====
frame 0000/0199 | err_norm=0.6665 | left_foot=0.0230 | right_foot=0.0245 | left_knee=0.0406 | right_knee=0.0875
frame 0010/0199 | err_norm=0.6510 | left_foot=0.0212 | right_foot=0.0228 | left_knee=0.0394 | right_knee=0.0860
frame 0020/0199 | err_norm=0.6473 | left_foot=0.0210 | right_foot=0.0218 | left_knee=0.0406 | right_knee=0.0857
frame 0030/0199 | err_norm=0.6393 | left_foot=0.0223 | right_foot=0.0213 | left_knee=0.0422 | right_knee=0.0832
frame 0040/0199 | err_norm=0.6288 | left_foot=0.0216 | right_foot=0.0181 | left_knee=0.0438 | right_knee=0.0830
frame 0050/0199 | err_norm=0.6312 | left_foot=0.0228 | right_foot=0.0179 | left_knee=0.0482 | right_knee=0.0924
frame 0060/0199 | err_norm=0.6440 | left_foot=0.0230 | right_foot=0.0128 | left_knee=0.0508 | right_knee=0.0925
frame 0070/0199 | err_norm=0.6476 | left_foot=0.0246 | right_foot=0.0122 | left_knee=0.0559 | right_knee=0.0923
frame 0080/0199 | err_norm=0.6579 | left_foot=0.0224 | right_foot=0.0071 | left_knee=0.0647 | right_knee=0.0910
frame 0090/0199 | err_norm=0.6666 | left_foot=0.0291 | right_foot=0.0075 | left_knee=0.0621 | right_knee=0.0900
frame 0100/0199 | err_norm=0.6513 | left_foot=0.0291 | right_foot=0.0091 | left_knee=0.0513 | right_knee=0.0869
frame 0110/0199 | err_norm=0.6544 | left_foot=0.0248 | right_foot=0.0111 | left_knee=0.0488 | right_knee=0.0857
frame 0120/0199 | err_norm=0.6691 | left_foot=0.0232 | right_foot=0.0117 | left_knee=0.0468 | right_knee=0.0860
frame 0130/0199 | err_norm=0.6536 | left_foot=0.0186 | right_foot=0.0114 | left_knee=0.0339 | right_knee=0.0859
frame 0140/0199 | err_norm=0.6433 | left_foot=0.0191 | right_foot=0.0172 | left_knee=0.0338 | right_knee=0.0879
frame 0150/0199 | err_norm=0.6477 | left_foot=0.0197 | right_foot=0.0159 | left_knee=0.0376 | right_knee=0.0919
frame 0160/0199 | err_norm=0.6479 | left_foot=0.0199 | right_foot=0.0144 | left_knee=0.0418 | right_knee=0.0930
frame 0170/0199 | err_norm=0.6483 | left_foot=0.0193 | right_foot=0.0137 | left_knee=0.0442 | right_knee=0.0928
frame 0180/0199 | err_norm=0.6533 | left_foot=0.0178 | right_foot=0.0164 | left_knee=0.0457 | right_knee=0.0937
frame 0190/0199 | err_norm=0.6678 | left_foot=0.0161 | right_foot=0.0123 | left_knee=0.0491 | right_knee=0.0928
frame 0199/0199 | err_norm=0.6782 | left_foot=0.0149 | right_foot=0.0110 | left_knee=0.0471 | right_knee=0.0927

=====
Retargeting summary
=====
frames: 200
mean final_error_norm: 0.6507701277732849
max final_error_norm: 0.6782450079917908
left_foot_err      mean=0.0217 max=0.0301
right_foot_err     mean=0.0146 max=0.0245
left_knee_err      mean=0.0462 max=0.0705
right_knee_err     mean=0.0890 max=0.0938
head_err           mean=0.0784 max=0.0926
chest_err          mean=0.0506 max=0.0601
hip_side_angle_deg mean=91.0280 max=96.9453
left_thigh_angle_deg mean=9.8983 max=10.9635
right_thigh_angle_deg mean=8.1451 max=9.2927
left_foot_forward_angle_deg mean=6.1101 max=38.6706
right_foot_forward_angle_deg mean=18.3329 max=28.1378
foot_sep           mean=0.1956 max=0.2962
target_foot_sep    mean=0.2042 max=0.3015

Saved R1 retargeted motion:
C:\Users\Mario\IsaacLab\scripts\r1\outputs\r1_retargeted_motion.npz
```

References

- Human2Humanoid:
<https://github.com/lecar-lab/human2humanoid>